

DW01_cmp2

2-Function Comparator

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Unsigned and signed (two's-complement) data operation

Revision History

Description

DW01_cmp2 is a two-input comparator. DW01_cmp2 compares two signed or unsigned numbers A and B and produces two output conditions LT_LE and GE_GT as results.

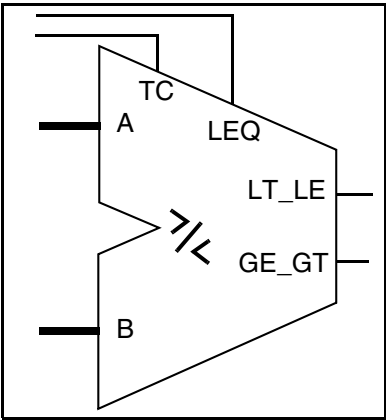


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>width</i> bits	Input	Input data
B	<i>width</i> bits	Input	Input data
LEQ	1 bit	Input	Output condition control
TC	1 bit	Input	Two's complement control ■ 0 = Unsigned ■ 1 = Signed
LT_LE	1 bit	Output	Less-than/less-than-or-equal output condition
GE_GT	1 bit	Output	Greater-than-or-equal/greater-than output condition

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 1	Word length of A and B

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
rpl	Ripple-carry synthesis model	none
pparch	Delay-optimized flexible parallel-prefix	DesignWare
apparch	Area-optimized flexible architecture that can be optimized for area, for speed, or for area, speed	DesignWare

a. During synthesis, Design Compiler will select the appropriate architecture for your constraints. However, you may force Design Compiler to use any architectures described in this table. For more, see [DesignWare Building Block IP User Guide](#).

Table 1-4 Simulation Models

Model	Function
DW01.DW01_CMP2_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_cmp2_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_cmp2.v	Verilog simulation model source code

Table 1-5 Functional Description

LEQ	Condition	LT_LE	GE_GT
1	$A \leq B$	1	0
1	$A > B$	0	1
0	$A < B$	1	0
0	$A \geq B$	0	1

The input signal LEQ determines whether the two output conditions are LT (less-than) and GE (greater-than-or-equal) ($LEQ = 0$) or LE (less-than-or-equal) and GT (greater-than) ($LEQ = 1$). The input TC determines whether the two inputs are compared as unsigned ($TC = 0$) or signed two's complement ($TC = 1$) numbers.

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Operator Inferencing - VHDL

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;

entity DW01_cmp2_oper is
    generic(wordlength: integer:=8);
    port(in1, in2      : in STD_LOGIC_VECTOR(wordlength-1 downto 0);
         instruction : in STD_LOGIC;
         comparison  : out boolean);
end DW01_cmp2_oper;

architecture oper of DW01_cmp2_oper is
    signal in1_signed, in2_signed: SIGNED(wordlength-1 downto 0);
begin
    in1_signed <= SIGNED(in1);
    in2_signed <= SIGNED(in2);
    -- infer the non-equality comparison operators
    process (in1_signed, in2_signed, instruction)
    begin
        if (instruction = '0') then
            comparison <= in1_signed > in2_signed;
        else
            comparison <= in1_signed >= in2_signed;
        end if;
    end process;
end oper;
```

HDL Usage Through Operator Inferencing - Verilog

```
module DW01_cmp2_oper(in1, in2, instruction, comparison);
    parameter wordlength = 8;

    input [wordlength-1:0] in1, in2;
    input instruction;
    output comparison;
    reg comparison;

    always @ (in1 or in2 or instruction)
    begin
        if (instruction == 0)
            comparison = (in1 > in2);
        else
            comparison = (in1 >= in2);
        end
    endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_cmp2_inst is
  generic ( inst_width : NATURAL := 8 );
  port ( inst_A : in std_logic_vector(inst_width-1 downto 0);
        inst_B : in std_logic_vector(inst_width-1 downto 0);
        inst_LEQ : in std_logic;
        inst_TC : in std_logic;
        LT_LE_inst : out std_logic;
        GE_GT_inst : out std_logic );
end DW01_cmp2_inst;

architecture inst of DW01_cmp2_inst is
begin

  -- Instance of DW01_cmp2
  U1 : DW01_cmp2
    generic map ( width => inst_width )
    port map ( A => inst_A, B => inst_B, LEQ => inst_LEQ,
              TC => inst_TC, LT_LE => LT_LE_inst, GE_GT => GE_GT_inst );
end inst;

-- pragma translate_off
configuration DW01_cmp2_inst_cfg_inst of DW01_cmp2_inst is
  for inst
  end for; -- inst
end DW01_cmp2_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_cmp2_inst( inst_A, inst_B, inst_LEQ, inst_TC,
                      LT_LE_inst, GE_GT_inst );

    parameter width = 8;

    input [width-1 : 0] inst_A;
    input [width-1 : 0] inst_B;
    input inst_LEQ;
    input inst_TC;
    output LT_LE_inst;
    output GE_GT_inst;

    // Instance of DW01_cmp2
    DW01_cmp2 #(width)
        U1 ( .A(inst_A), .B(inst_B), .LEQ(inst_LEQ), .TC(inst_TC),
            .LT_LE(LT_LE_inst), .GE_GT(GE_GT_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed Table 1-4, “Obsolete Synthesis Implementations”
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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